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The evolution of cooperation in asymmetric systems

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Explaining "Tragedy of the Commons" of evolution of cooperation remains one of the greatest problems for both biology and social science. Asymmetrical interaction, which is one of the most important characteristics of cooperative system, has not been sufficiently considered in the existing models of the evolution of cooperation. Considering the inequality in the number and payoff between the cooperative actors and recipients in cooperation systems, discriminative density-dependent interference competition will occur in limited dispersal systems. Our model and simulation show that the local but not the global stability of a cooperative interaction can be maintained if the utilization of common resource remains unsaturated, which can be achieved by density-dependent restraint or competition among the cooperative actors. More intense density dependent interference competition among the cooperative actors and the ready availability of the common resource, with a higher intrinsic contribution ratio of a cooperative actor to the recipient, will increase the probability of cooperation. The cooperation between the recipient and the cooperative actors can be transformed into conflict and, it oscillates chaotically with variations of the affecting factors under different environmental or ecological conditions. The higher initial relatedness (i.e. similar to kin or reciprocity relatedness), which is equivalent to intrinsic contribution ratio of a cooperative actor to the recipient, can be selected for by penalizing less cooperative or cheating actors but rewarding cooperative individuals in asymmetric systems. The initial relatedness is a pivot but not the aim of evolution of cooperation. This explains well the direct conflict observed in almost all cooperative systems.

cooperation, altruism, density dependent, tragedy of the commons, oscillation, chaos, asymmetric interaction, interference competition

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Explaining how the stability of cooperative systems is maintained has been one of the central problems in both biology and social science since Charles Darwin and Adam Smith [1–4]. Popular theories have shown that the initial relatedness coefficient between the involved partners is sufficient to maintain cooperative interaction, and that this initial relatedness coefficient between the partners is created by genetic similarity [5–7], mutual reciprocity [8–10], intergroup competition [11,12], or any other mechanism that creates a positive fitness correlation [2]. However, an increase in the cooperative actors, which has a positive feedback on the recipient in a cooperative interaction, will also increase their competition with the recipient if the local resources or space are limited. The cooperative actors must compete with the recipient after the common resource is saturated, which is a fundamental theoretical dilemma, the "tragedy of the commons" [13–17].

That direct conflict (competition) will occur after a common resource is saturated after an increase in the involved partners, and will disrupt a cooperative interaction,

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was recognized by Hardin in 1968, William in 1976, and subsequently by other researchers, including when there is a strong genetic or reciprocal relatedness between the partners [18–21]. Hamilton and others have suggested that self restraint or spatial heterogeneity (restraint) will prevent direct conflict between the cooperative actors and the recipient that results from the increase in the partners [7,9,22–25]. However, these theories have drawn several criticisms. (i) The cost of self restraint, which is based on the "iterated prisoner's dilemma" through "tit-for-tat", will increase with the augmentation of the cooperative actors or in further repetitions of the game. Therefore, the extra cooperative actors or the extra repeated actions of the cooperative actors might not result in a reward from the recipient after the common resource is saturated [24,26]. (ii) Spatial heterogeneity (constraint), which is closely related to self restraint, is unable to readily explain why mutants that overcome the spatial restraint are not copied in the next generation, because competitive individuals who overcome the spatial constraint will increase in fitness when the space is saturated [26–28]. When multiple players are considered rather than two players, the spatial constraint will often inhibit the evolution of cooperation because the competition among the cooperative actors will increase if the spatial constraint creates a local space for them, and the less-altruistic individuals or non-cooperating actors will have greater fitness [27,28]. (iii) It has been shown that cooperative actors (e.g., subordinates of intra-specific cooperation, symbionts of mutualism) can use non-cooperating strategies to compete with the queen or host for reproduction in both eusociality (e.g., bee, ant, or mole rodent) and reciprocal mutualism (e.g., fig-fig wasp, yuccas-yuccas moth mutualism), and that these cooperative systems occur with high genetic similarity or mutual reciprocity, implying that genetic similarity or mutual reciprocity alone is not sufficient to maintain the stability of cooperation [29–33].

These main models of the evolution of cooperation, including Hamilton's rule, Iterated Prisoners' Dilemma, and group selection, are essentially based on the assumption that cooperative actors and the recipient constitute a mathematically symmetric interaction. However, the empirical data/observations have shown that there exist asymmetric interactions within cooperative interactions [5,8,9,12,34,35]. In realistic cooperative systems, the cooperative actors and recipient usually form an asymmetric interaction [21,30,35–38]. This asymmetry exists in three forms: (i) Inequality in payoff, e.g., the benefit to the dominant individual or hosted species in a cooperative system will be Greater than the benefit to subordinates or symbionts [17,34,35]; (ii) the evolutionary paths or contribution pathways to each other differ, e.g., the evolutionary velocity of symbionts is higher than that of the hosted species in reciprocal mutualism [29], and the dominant individuals might implement a strategy to punish non-cooperating subordinates and reward cooperating subordinates, while the subordinates implement a strategy of either paying the cost of cooperating or of not cooperating with the dominant individual [30,33,37]; (iii) a discrepancy in the information state between the recipient and the cooperative actors, in that the hosted species or the dominant individual (recipient) might not always be aware which of the symbionts or subordinates (cooperative actors) is non-cooperative, and the symbionts or subordinates might also be incompletely aware when and how intense the punishment or reward conferred by the hosted species or dominant individual will be if they undertake a strategy of non-cooperation [35,37,39,40]. These asymmetric interactions are largely attributable to the fact that the number of cooperative actors is usually larger than the number of recipients in cooperation systems, or that the pay-off of a cooperative actor is lower than the recipient in the game of cooperation. Here, we would like to model how stable cooperation is maintained by considering the discriminative density-dependent interference competition that results from the inequality in numbers and payoff of cooperative actors and recipients.

1 Conceptual review

In the classical models of the evolution of cooperation, the fitness of the recipient and the cooperative actors is a linear function of other non-genetic or non-exchanged factors. Therefore, the regression coefficient between the fitness of the recipient and that of the cooperative actors is independent of environmental or ecological factors. This basic presumption has been specifically clarified by various investigation [6–10,41–43]. These models can be written simply as:

$$w_i = b_{0i} + \sum_{j=1}^{N} r_{ij} a_{ij} \tag{1}$$

If all the contributions of neighbors are equal, the equation reduces to $w=b_0+raN=b_0+bN$, where $b=ra$, which indicates that fitness w is a linear function of the number of cooperative actors (Figure 1A). The fitness correlation between the recipient and the number of cooperative actors is therefore independent of other non-genetic or non-exchanged factors, and is mainly determined by their genetic similarity, degree of reciprocity, or other similar mechanisms.

In these simplified models, although the positive contribution of the cooperative actors to the recipient is created by different mechanisms of kin selection, reciprocity selection and group selection, the mathematical process that creates the positive fitness regression is similar in the different selection theories [10,42,44–46,47]. This positive fitness regression coefficient can also be created by other factors in cooperative interaction [2]. In the above models, the fitness regression coefficient between the recipient and cooperative actors is independent from the number of the cooperative actors, which neglects the density dependence effect (i. e. fitness relatedness will be equal to the genetic or reciprocity relatedness, see appendix).

However, if the availability of the common resource or space is limited, the marginal values for the fitness of both the recipient and the cooperative actors will be a function of the reduction in the availability of the common resource or space [2,18,19]. An increase in the cooperative actors will lead to a reduction in the fitness of the recipient after the common resource is saturated (Figure 1B). Direct conflict between the cooperative actors and the recipient will occur with an increase in the number of cooperative actors if the availability of the common resource or space is limited [14,18, 19,21,48]. Hamilton himself and other scientists argued that spatial constraints created by recipient or self restraint of cooperative actor will play a role in preventing conflicts that results from an increase in the number of cooperative actors [7,22,24]. Figure 1 (C and D) describes the fitness pattern between the cooperative actors and the recipient, given that the spatial constraint created by the recipient or the self restraint of the cooperative actors is sufficient to prevent direct conflict.

2 Parameter assumptions

In this model, we make 4 basic assumptions. (i) There exists a barrier to the free dispersal of the involved partners to another colony, and the number of cooperative actors is larger than the number of recipients. The sum of the payoff for multiple cooperative actors utilizing a common resource or space is therefore roughly matched with the pay-off for one recipient. (ii) The contribution of the cooperative actors to the recipient (i.e., the fitness of the recipient) and the utilization of a common resource at the expense of the fitness of the recipient (i.e. the cost to the recipient) occur during the same period. (iii) The common resource or space available to the involved partners is limited. Therefore, an increase in the fitness of either the cooperative actors or of the recipient has a marginal effect. (iv) Because of the barrier to dispersal, the density-dependent interference competition among the cooperative actors is higher than with recipient. Such interference competition might reduce both the contribution of the cooperative actors to the recipient and the utilization of the common resource by the cooperative actors themselves. Based on these assumptions, we model the fitness of the

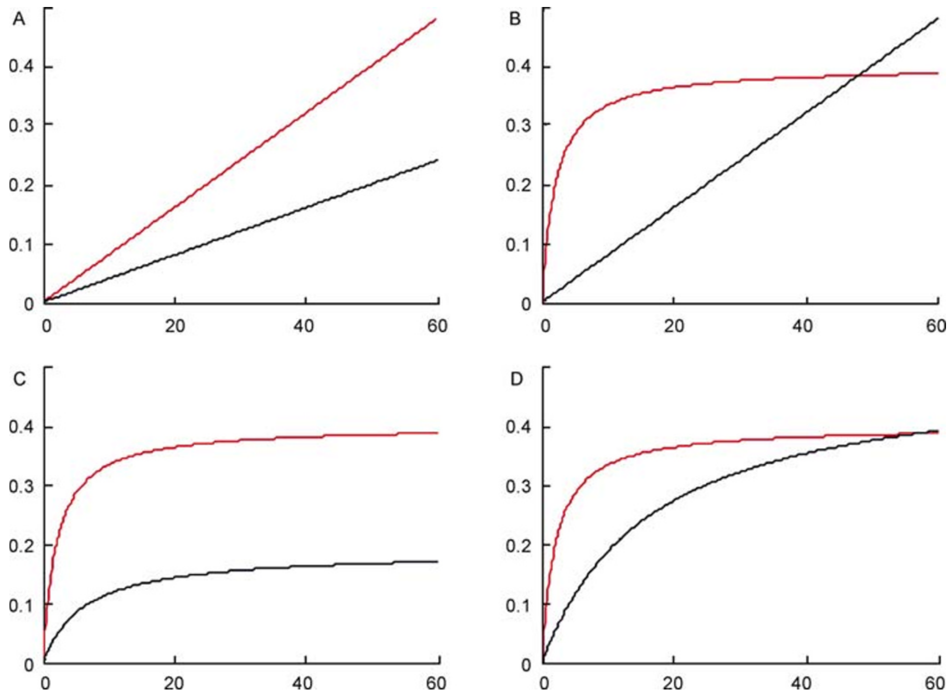

Figure 1 The benefit (B) and cost (C) as a function of the number of cooperative actors. If (B) > (C) the net benefit will be greater than zero, and cooperation is maintained, whereas if (B) < (C), the net benefit of the cooperation will be less than zero and conflict will occur, disrupting the cooperative interaction. A, When α and β in equation (2) are zero, the functions of both the benefit and the cost will be linear functions, as $b(N)+b_0$ and $c(N)+c_0$. Here b=0.008, c=0.004, $\alpha_1=\alpha_2=0$, and $\beta_1=\beta_2=0$. B, When β_1 is zero in the benefit function, the function of the benefit is expressed as $\frac{bN+b_0}{1+\alpha_1 N}$, whereas the function of the cost is $c(N)+c_0$; here b=0.20, c=0.008, α_1=0.5, β_1=0, α_2=0, and β_2=0. (C and D) When β is zero in both the benefit and cost functions, and the benefit and cost are expressed as $\frac{bN+b_0}{1+\alpha_1 N}$ and $\frac{cN+c_0}{1+\alpha_2 N}$, then b=0.2, c=0.03, α_1=0.5, β_1=0, α_2=0.16, and β_2=0. C, The common space or resource for the benefit is larger than the cost. D, The space and resource for the cost and the benefit are equal; here b=0.2, c=0.03, α_1=0.5, β_1=0, α_2=0.06, and β_2=0. Red, benefit; black, cost.

recipient as:

The net benefit of recipient $B(net)=B(N)-C(N)$, where

$$B(N)=\frac{bN+b_0}{1+\alpha_1 N+\beta_1(N-1)^2},$$

$$C(N)=\frac{cN+c_0}{1+\alpha_2 N+\beta_2(N-1)^2}, \qquad (2)$$

In this function, $B(N)$ is transformed to a linear function of N described in Figure 1A when $\alpha_1=\beta_1=0$. The net benefit to the recipient may decrease, becoming negative, with an increase in the number of cooperative actors (Figure 1B), if $\beta_1=0$ and the cost is a linear function of the number of cooperative actors. However, if there exists spatial heterogeneity (restraint), which prevents overexploitation by the extra cooperative actors, the recipient and the cooperative actors attain a stable equilibrium with the fitness pattern described as shown in Figure 1C and D when the density-dependent interference competition $\beta=0$ [24]. If density-dependent interference competition β exists among the cooperative actors, the fitness pattern of the recipient will be described as shown in Figure 2. We would like to note here that the parameter of the number of cooperative actors is replaceable by any other variables in both empirical data analysis and theoretical models, e.g. investment of cooperation behavior, time, or any others.

At $N=1$, $B(N)=B(1)=(b+b_0)/(1+\alpha_1)$, where $B(1)$ is, in fact, the initial fitness growth rate of the recipient, considering the contribution of one cooperative actor. This is treated as the intrinsic growth rate of the gross benefit, denoted by r_m. The parameter α_1 is described as $\alpha_1=\frac{b+b_0}{r_m}-1$. α_1 is the parameter that describes the intrinsic growth rate [48]. When α_1 is smaller, the intrinsic growth rate will be larger. The explanation of $C(N)$ is similar to that of $B(N)$.

The intrinsic contribution ratio of a cooperative actor to recipient, which is equivalent to the initial value of fitness regression coefficient between the recipient and cooperative actors at $N=1$, will be equal to genetic or reciprocity relatedness, assuming that it is a linear evolution process from the genotype to the fitness advantage. For simplicity of calculation, an assumption of $b_0=c_0=0$ is made in the following discussion, calculating the intrinsic contribution ratio of cooperative actor to recipient without density effect as $R=\frac{B(1)}{C(1)}=\frac{b}{c}(\frac{1+\alpha_2}{1+\alpha_1})=\frac{b}{c}K^*$, where $K^*=\frac{1+\alpha_2}{1+\alpha_1}=1+\frac{\alpha_2-\alpha_1}{1+\alpha_1}$. Given $\frac{b}{c}$, the intrinsic contribution ratio of a cooperative actor to recipient R depends on the ratio of $1+\alpha_1$ to $1+\alpha_2$. This intrinsic contribution ratio is mainly

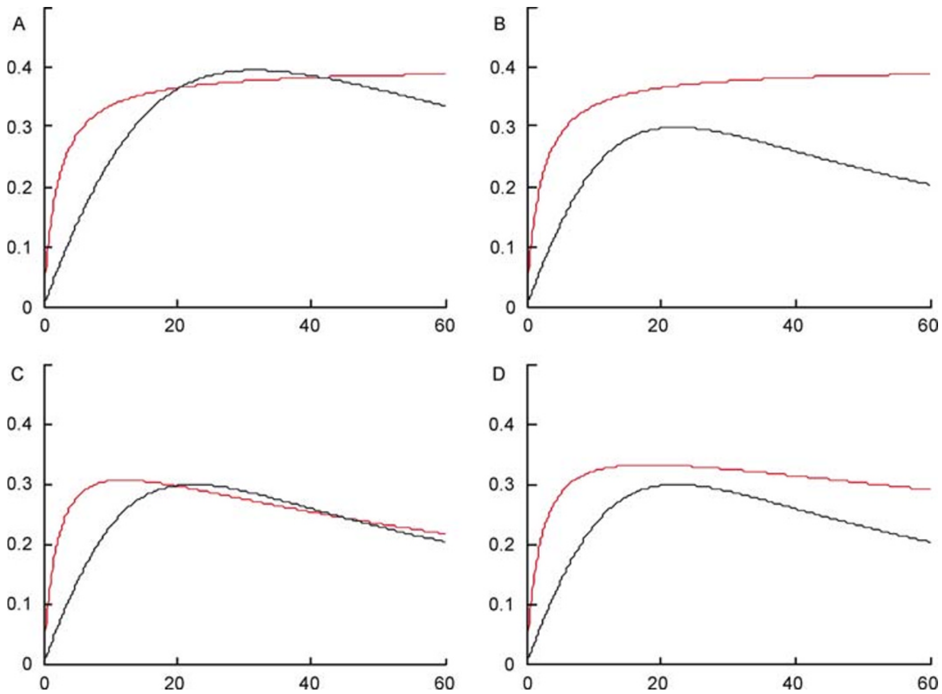

Figure 2 The benefit (B) and cost (C) as a function of the number of cooperative actors, when density-dependent interference competition exists among the cooperative actors. The parameters α and β are not zero in Function 2. These figures indicate that the net benefit is either possibly lower or always higher than zero, if density-dependent interference competition exists among the cooperative actors. A and B, Density-dependent interference competition among the cooperative actors only exists in a common resource utilization by the cooperative actors (cost to the recipient), while the other parameters are held constant. A, $b=0.2$, $c=0.03$, $\alpha_1=0.5$, $\beta_1=0$, $\alpha_2=0.015$, $\beta_2=0.001$. B, $b=0.2$, $c=0.03$, $\alpha_1=0.5$, $\beta_1=0$, $\alpha_2=0.015$, $\beta_2=0.002$. C and D, Density-dependent interference competition among the cooperative actors exists in both a common resource utilization by the cooperative actors (cost to the recipient) and the contribution of the cooperative actors to the recipient (benefit to the recipient), which changes the value of the density-dependent restraint. C, $b=0.2$, $c=0.03$, $\alpha_1=0.5$, $\beta_1=0.007$, $\alpha_2=0.015$, and $\beta_2=0.002$. D, $b=0.2$, $c=0.03$, $\alpha_1=0.5$, $\beta_1=0.003$, $\alpha_2=0.015$, and $\beta_2=0.002$. Red, benefit; black, cost.

determined by genetic factors or mutual reciprocity, because the intrinsic growth rate, without considering environmental or behavioral constraints, is mainly determined by the intrinsic characteristics of the involved partners.

β is the parameter that influences fitness or cost through the density of the cooperative actors. The greater the number of cooperative actors, the greater will be the negative effect of β on the benefit and cost growth rates. Theoretically, such density-dependent factors constitute either behavioral competition or negative interference among the cooperative actors, which will be more intense as their numbers increase, or density-dependent regulation, in which the individuals involved actively regulate (self restraint) the increase in the ratio as a function of their own number. Both mechanisms will reduce the fitness growth rate of each of the individuals involved.

3 Functional response and the dynamics of cooperation

The parameters α and β determine the pattern of variation in the fitness of the recipient. Parameter α describes the intrinsic contribution rate of the cooperative actors to the recipient or the reward rate of the recipient to the cooperative actors. When the parameter β=0 and there exists spatial heterogeneity (restraint), the fitness pattern of $B(N)$ and $C(N)$ will increase asymptotically as a function of the number of cooperative actors. The growth rate is determined by the parameter α (Figure 1C and D). However, when $b\neq0$ and there is no spatial heterogeneity (no restraint), the gross fitness of the recipient or the cost will be affected by the density of the cooperative actors. Given the effect of density-dependent interference competition as a function of the number of cooperative actors, the gross benefit to the recipient $B(N)$ might either be higher or lower than the cost $C(N)$, depending on the parameters α and β (Figure 2).

We can solve the equation $B(net)=B(N)-C(N)=0$ to find an equilibrium solution, which leads to:

$$(b-c)+(b\alpha_2-c\alpha_1)N+(b\beta_2-c\beta_1)(N-1)^2=0\,. \qquad (3)$$

Without loss of generality, let $\alpha_1=\alpha$, $\alpha_2=k\alpha$, $\beta_1=\beta$, and $\beta_2=m\beta$, then the above equation becomes:

$$\alpha=\frac{(b-c)+(bm-c)(N-1)^2\beta}{(c-bk)N}\,. \qquad (4)$$

In a given specific cooperative system, the difference (e.g., parameters m and k) between the contribution of the cooperative actors to the recipient (e.g., benefit to the recipient) and the common resource utilization by the cooperative actors themselves (e.g., cost to the recipient) are given in terms of both the intrinsic growth ability α and the interference competition b. These biological characteristics of the parameters m and k might be the intrinsic characteristics of cooperative systems. Because both the intrinsic growth ability α and the interference competition β vary in different environments, and the trade-off between the recipient and the cooperative actors might vary in different environments, considering the effect of the number of cooperative actors (Figure 3).

However, if b, c, α,β are given, the equation $B(N)-C(N)=0$ is solved as

$$m=(\frac{c}{b})+(\frac{\alpha}{\beta})(\frac{c}{b}-k)\frac{N}{(N-1)^2}-\frac{(1-c/b)}{\beta(N-1)^2}\,. \qquad (5)$$

The differences (e.g., parameters m and k) in both the intrinsic growth ability α and the interference competition β vary in different systems. A stronger asymmetric interaction between the recipient and the cooperative actors might create a larger difference in parameters m and k. This is because a stronger asymmetric interaction favors sanction of a dominant individual or hosted species, dictating that the cooperative actors contribute more or reduce their utilization rate of the common resource. This discriminative selection of the intrinsic growth ability and the discriminative effect of interference competition in fitness increase between cooperative actors and recipient will favor the evolution of cooperation (Figure 4).

4 Trade-off for the recipient

In an asymmetric cooperative system, if density-dependent interference competition exists among the cooperative actors, the parameter of interference competition that reduces the utilization of a common resource or space, as well as the intrinsic contribution ratio of a cooperative actor to recipient, will be crucial in the fitness interaction between the cooperative actors and the recipient (Figure 2). When $B(net)>0$, a

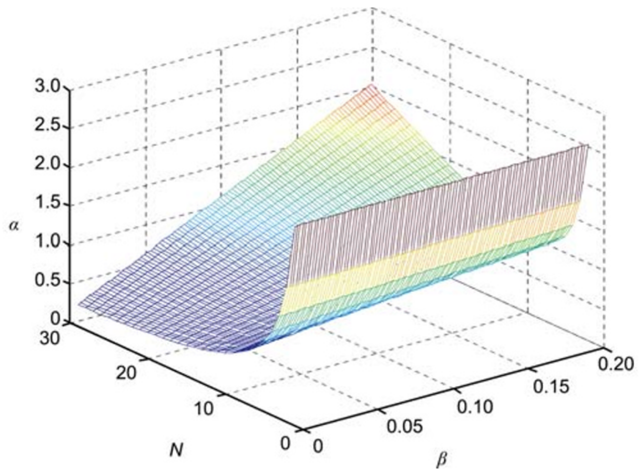

Figure 3 The relationship between the intrinsic growth rate ability α, the interference competition β, and the number of cooperative actors N at the point at which the benefit to the recipient is equal to the cost (reward to the cooperative actors). Here, b=0.4, c=0.1, k=0.1, and m=0.3.

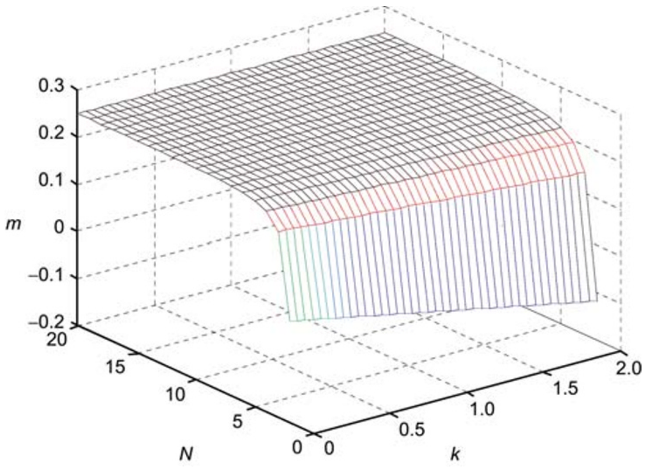

Figure 4 The relationship between the ratio k of the intrinsic growth ability of the benefit B and the cost C, the ratio m of the interference competition of the benefit B and the cost C, and the number of cooperative actors N at the point at which the benefit to the recipient is equal to the cost (reward to the cooperative actors). Here, b=0.4, c=0.1, α=0.1, and β=3.

cooperative strategy undertaken by the cooperative actors will achieve higher fitness than will spite. However, when $B(net)<0$, cooperating with the cooperative actors will cause a net loss of benefit for the recipient. The cooperative interaction cannot be maintained when $B(net)<0$, but will be transformed into a conflict interaction [24,47]. The net benefit to the recipient is calculated with the following function:

$$B_{net} = \frac{bN + b_0}{1+\alpha_1 N + \beta_1 (N-1)^2} - \frac{cN + c_0}{1+\alpha_2 N + \beta_2 (N-1)^2}. \quad (6)$$

5 Local stability of cooperative interaction

Our simulation showed that discriminative intrinsic commons utilization (i.e., the intrinsic growth rate) and density-dependent interference competition between the recipient and the cooperative actors will directly determine whether or not the net benefit to the recipient will be main

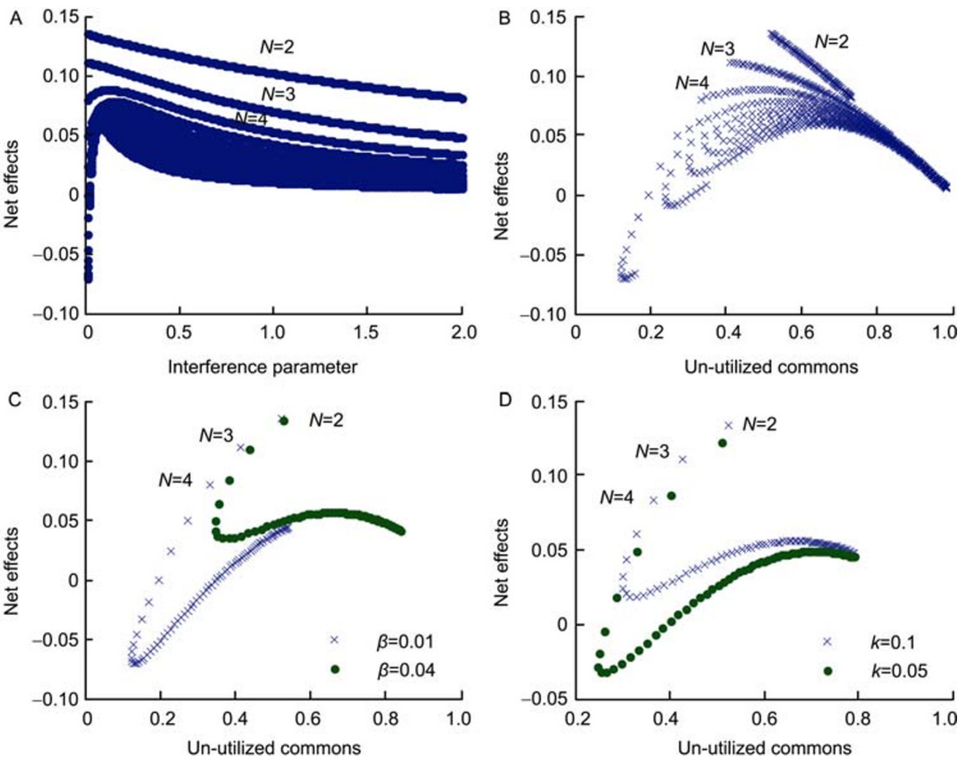

Figure 5 The net benefit to the recipient in different situations. The simulations show that a positive net benefit to the recipient might transform into a negative net benefit with variation in the interference competition β, the intrinsic growth ability α, or the availability of an unutilized common resource [1−$B(N)$−$C(N)$]. A, Net benefit versus interference competition parameters at α=0.8, k=0.1, and m=0.5, indicating that a negative net benefit might only occur when the density-dependent interference competition β is weak. B, Net benefit versus the availability of an unutilized common resource at α=0.8, β=0.01, k=0.1, and m=0.5, indicating that the negative net benefit might occur when the availability of the unutilized common resource is very low. C, Net benefit versus an unutilized common resource at different β, while the other parameters remain the same, with α=0.8, k=0.1, and m=0.5, indicating that variation in the density-dependent interference competition β might lead to different trade-offs for the recipient. D, Net benefit versus an unutilized common resource with differences in the intrinsic growth ability in the benefit and cost to the recipient, where α=0.8, β=0.01, and m=0.5, indicating discrimination selection against the cooperative actors for a higher contribution to the recipient, whereas less efficient utilization of the common resource will favor the maintenance of the cooperative interaction. Here, b=0.4, c=0.1, and the number of cooperative actors varied from 2 to 60.

tained above zero. Given that the effect of density-dependent interference competition for the recipient does not exist, the net benefit to the recipient is maintained above zero when the density-dependent interference competition among the cooperative actors for the commons utilization is large (Figures 5C and D). A low intrinsic growth rate of cooperative actors (i. e., commons utilization rate of cooperative actors), will also favor the maintenance of the net benefit above zero (Figure 5D). However, this maintenance above zero is vulnerable. The net benefit may decrease to below zero if the common resource availability, the intrinsic growth ability α, or the density-dependent interference competition β changes (Figure 5B–D). This possibility exists under natural conditions, because the density-dependent interference competition parameter β, the intrinsic growth ability α, and the common resource or space change in different ecological or environmental situations. A change in either the intrinsic growth ability α or the density-dependent interference competition parameter β might lead to a net benefit for the recipient of less than zero, and a competitive (conflict) interaction will occur after the common resource or space is saturated. A competitive interaction will be created when the availability of the common resource or space is shortened, given the growth rate cycle α and the density-dependent interference competition parameter β.

The intrinsic growth ratio between the gross benefit and cost, which is described by the growth rate ability α, will-greatly affect the distribution pattern of the cooperative or conflict interaction. This ratio (i. e., intrinsic contribution ratio of cooperative actor to the recipient) between the intrinsic gross benefit and cost $R=B/C$ is equivalent to the reciprocity relatedness or kin relatedness [2,10] (appendix). Our simulation showed that higher initial relatedness (i.e., similar to genetic relatedness or reciprocal relatedness) increases the probability of a cooperative interaction, and therefore reduces the probability of a conflict interaction (Figure 6). Such high initial relatedness (i. e., intrinsic contribution ratio of cooperative actor to the recipient) will favor the evolution of a cooperative interaction, but it is not sufficient to maintain the global stability of the cooperative interaction.

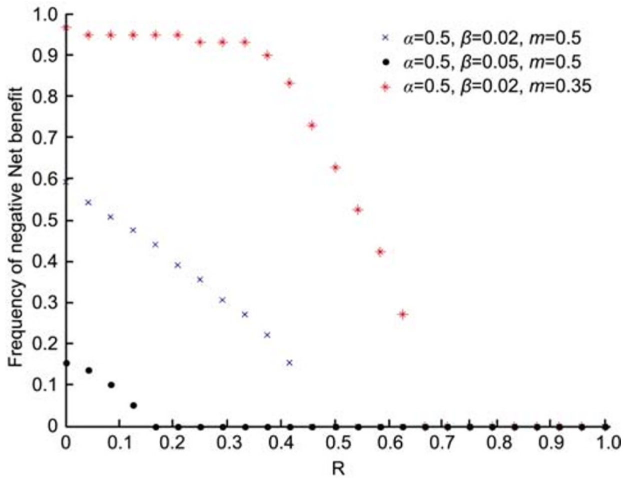

Figure 6 The frequency of the negative benefit at different levels of initial relatedness between the cooperative actors and the recipient (note: in asymmetric systems, such initial relatedness might be created through the discriminative sanction of recipient against the non-cooperative or less cooperative individuals of cooperative actors while rewarding the cooperative individuals, see Wang *et al*. manuscript accepted for publication in *Ecology*; Rui-Wu Wang *et al*. unpublished model).The figure shows that a higher initial correlation coefficient between the cooperative actors and the recipient reduces the probability of a negative benefit. The number of cooperative actors varied from 2 to 60, b=0.4, and c=0.1. Here, the initial relatedness $R=(B)/(C)$ at N=1, which is equivalent to the genetic or reciprocal relatedness (appendix), is standardized in the figure, with k varying from 0.0001 to c/b.

6 Discussion

In the original dominant theories of the evolution of cooperation, it was assumed that no marginal effect was exerted on either the cost or the benefit by extra cooperative actors or further repeated cooperative actions, and that the game of cooperation is for two players rather than for multiple players. A cooperative equilibrium evolves when the genetic or reciprocal relatedness is high. However, the later development of these dominant theories demonstrated that a stable cooperation equilibrium is reached through self-restraint, spatial heterogeneity, or positive assortment, when the marginal effect is considered [5,8–10,49,50]. This cooperation equilibrium is only maintained if the cooperative partners undertake a pure cooperating strategy [2,10,25,50–53].

However, such a pure cooperative strategy may not be credible if multiple players and the marginal (density-dependent) effect of their benefit or cost are considered in the game of cooperation [27,28,52,53]. A cheating or non-cooperating strategy will confer a fitness advantage after common resource is saturated or if the cost of cooperation is greater than the benefit, and therefore a cooperative Nash equilibrium cannot evolve [26,52,53]. Empirical observations have further demonstrated that cooperative actors undertake mixed strategies, either cooperating with the recipient or cheating on the recipient, in both inter-specific mutualism and intra-specific cooperation [29,31,33,37,54]. Direct conflict between cooperative actors and the recipient has also been widely observed in most cooperative systems [26,30,35,55].

If the effect of density-dependent interference competition is considered, although it has been widely demonstrated in population biology, the utilization of a common resource will remain unsaturated, and the local cooperative interaction will thus be maintained. Because the number of cooperative actors is usually larger than the number of recipients in evolutionary stable cooperation systems, discriminative density-dependent interference competition between the cooperative actors and recipients could occur. An increase in the interference competition among the cooperative actors occurs when the number of cooperative actors increases because a common resource or space is limited. This will reduce the efficiency of the utilization of a common resource by the involved partners, thereby keeping a common resource or space unsaturated. In this situation, the positive net benefit to the recipient is maintained. If the common resource remains unsaturated, the conflict or competition for a common resource, which could disrupt the cooperative system, will not occur.

Such discriminative density dependent interference competition among cooperative actors might vary at different age developments, food or any other commons availabilities. In eu-sociality systems of bees, ants, and wasps, the workers presented strong mutual policing, but not for the queen. The intensity of such policing varies greatly at different conditions (e. g. chemically mimicry of eggs) or between different species [33]. Such mutual policing between workers will reduce potential conflicts for reproduction between workers and the queen [33]. Buss [56] has experimentally demonstrated that density dependent interference competition among individuals of a group living bryozoan (*Bugula turret*) strongly depend on their population size, and some individuals might loss their propagation in high density, which will reduce commons competition between group living individuals. Interference competition discrimination also exists between males and females of meerkats and the interference competition within male and female individuals varies greatly at different development periods [57]. In inter-specific cooperation, Wang *et al.* [58] directly reported that interference competition among the symbionts (fig wasps) keeps the commons utilization unsaturated, and therefore maintains cooperative interaction between reciprocal mutualists in fig/fig wasp mutualism, whereas such density dependence interference competition among fig wasps obviously differs at different seasons because of the different lifespans of pollinator wasps at different temperatures and humidity [58,59,60].

That the effect of density dependent interference competition varies under different ecological or environmental conditions might lead that cooperative interaction to transform into a conflict interaction, because the net benefit is sensitive to parameters of intrinsic growth ability α and intensity of interference competition β. Both the intrinsic growth ability α and density-dependent interference competition β vary greatly in real cooperative systems in different environments. The intrinsic growth rate described by α increases when living conditions change for the better, whereas it greatly decreases when living conditions worsen. The plasticity of the intrinsic growth rate ranges broadly in many organisms. Given the same density-dependent interference competition, a higher intrinsic growth rate of the cooperative actors will more easily lead to direct fitness conflict (Figure 3D).

Like the intrinsic growth rate, density-dependent interference competition, which is possibly determined by the availability of a common space or resource, also varies greatly under different natural conditions. Density-dependent interference competition will be weakened if the individuals involved in the cooperative interaction enter the system sequentially, or if the generations of cooperative actors do not overlap significantly. The availability of a common resource or space to the recipient and cooperative actors, which varies greatly under different natural conditions and might simultaneously lead to variations in parameters α and β, also leads to different trade-offs between the cooperative actors and the recipient.

Because the factors that determine the trade-off between the cooperative actors and the recipient vary greatly under natural conditions, direct competition between the recipient and the cooperative actors may exist in cooperative systems. This explains why direct competition is sometimes observed between the cooperative actors and the recipient in almost all well-documented cooperative systems [26,30,33,37,39]. This analysis also explains why the competition among the involved partners of cooperative systems varies in different situations in many empirical reports [26,37,55,59,61].

When cooperative actors compete with the recipient, taking a non-cooperating strategy should be the predominant strategy for the cooperative actors. Thus, honest cooperative actors may transform into non-cooperative individuals, parasitizing the cooperative system. Therefore, symbionts or subordinates (cooperative actors) undertake a mixed strategy, being either cooperative actors or non-cooperative actors that parasitize the cooperative system. This has been widely observed in both intra-specific and inter-specific cooperation interactions [26,29,30,33,37,39].

Although competition has been observed in most cooperative systems, how is a competitive interaction depressed to maintain the cooperative interaction and who pays the extra cost to depress the competition? Our model implies that the depression of a utilization of the common resource by reducing the intrinsic growth rate, reducing the efficiency of a utilization of the common resource or space (through density-dependent interference competition), or reducing the increase in the number of cooperative actors favors the maintenance of a cooperative interaction between the cooperative actors and the recipient. In an asymmetric cooperative system, the extra cost required to depress the increase in cooperative actor numbers or the efficiency of common resource or space utilization might be paid by the hosted species or by the dominant individual, which owns a commons resource or space. The payoff for the hosted species or the dominant individual will be much higher than that for the subordinate or symbiont, when cooperative interaction is maintained in asymmetric cooperation. This is a "boxed pigs", in which the player who gains a much higher benefit will definitely pay the extra cost to do the public service of maintaining the cooperation interaction, whereas the player who gains less benefit will definitely not pay the cost to do such public service. In such a payoff matrix, both players achieve an evolutionarily stable strategy [10].

Theoretically, the density-dependent negative effect on the increase in either the cost or benefit to the recipient could be both a passive interference competition among the cooperative actors and a positive density-dependent regulation of the cooperative actors, but the passive interference competition between the cooperative actors seems more credible. This is because a greater number of cooperative actors will lead to a reduction in a common resource or space. Then competition or interference among the cooperative actors might occur if a common resource or space is limited and there is a barrier to the dispersal of the involved partners.

However, the density-dependent regulation (self restraint) arising from genetic similarity or mutual reciprocity between the cooperative actors and the recipient might not be credible [24]. An increase in the cooperative actors will lead to less-available common resource or space, and the extra cooperative actors or repeated cooperative actions will not receive any fitness advantage if the self-restraint strategy is undertaken after the common resource or space is saturated. Such density-dependent mechanisms therefore might be passive actions but not positive actions in cooperation systems. This passive density-dependent interference competition among cooperative actors that maintains a cooperation interaction might be a general mechanism, rather than a specific mechanism only applicable to specific cooperation systems [26,56,62].

7 Conclusions

Our results indicate that density-dependent interference competition among cooperative actors can maintains the local stability of a cooperative interaction in asymmetric systems. The probability of cooperation or competition between the cooperative actors and the recipient will be determined by the intrinsic growth rates of both the cost and the benefit to the recipient, the density-dependent interference competition, and the availability of a common resource or space. Changes in any of these parameters, individually or together, might lead to different fitness interaction between the cooperative actors and the recipient. Because these factors that determine the fitness pattern or trade-off between the cooperative actors and recipient vary in different environments [63,64], the fitness interaction between the cooperative actors and recipient differ with changes in the environment.

The initial relatedness, which is the genetic or reciprocal relatedness in kin or reciprocity selection theories, but might be the contribution ratio of a cooperative actor to recipient in asymmetric cooperation systems through the enforcement of recipient, will favor the evolution of a cooperative interaction, and this is the pivot but not the aim of the evolution of cooperation [65]. Density-dependent interference competition only maintains the local stability of a cooperative interaction. Therefore, the cooperative interaction will oscillate chaotically with variations in the intrinsic growth rate of both the cost and the benefit to the recipient, the density-dependent interference competition, and the availability of a common resource or space [22,66]. Given that direct competition exists in cooperative systems, non-cooperating behavior and mutants that are cheaters and parasitize a cooperative system are possible. This explains why direct competition or conflict between the cooperative actors and the recipient has been observed in most cooperative systems, and why cheaters have also been observed among the subordinates or symbionts of both intra-specific and inter-specific cooperative systems.

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Appendix: The relationship between fitness relatedness or the cost/benefit ratio at N=1 and the genetic or reciprocity relatedness.

If the evolution of players from their genotype to their fitness is a linear process, the fitness correlation (relatedness) between the recipient and the cooperative actors will be equal to their genetic relatedness. This logic is directly implied in Queller's equation for the evolution of cooperation [42]. Figure 7 illustrates the path diagram of the linear evolution of cooperation, adapted from the figure of Queller. The mathematical demonstration follows.

Suppose that w_1 and w_2 are the fitnesses of two players

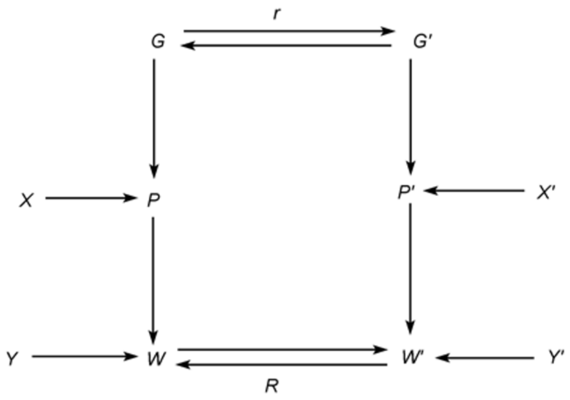

Figure 7 Path diagram for the effect of breeding values of individuals (G) and of their partners (G') on fitness (W), by way of phenotypes (P and P'). X and X' are additional factors affecting phenotypes, with Y representing any additional factors affecting fitness. Here X, X', and Y are shown to be uncorrelated with breeding values, but this will not always be true. Although path analysis uses standardized variables, the variables used in this are unstandardized.

with respectively genotypes of g_1 and g_2. In our system, $w_1=B(N)$. $C(N)$ is the cost of player 1, which is transformed into the fitness of player 2 with a constant multiplier, i.e. $w_2=kC(N)$, where k is a constant, when N=1, and there is no density effect to be considered. Assume that w_1 and w_2 are linearly related to their genotypes g_1 and g_2, then according to Fisher's rule or the Price equation, $dw_1=c_1dg_1$ and $dw_2=c_2dg_2$, so

$$\frac{dw_1}{dw_2}=c\frac{dg_1}{dg_2}=cr$$

where r is the genetic relatedness of the two players and $c=c_1/c_2$ is a constant. This gives

$$w_1=a_1w_2r+a_0$$

If w_1=0, then w_2= 0, which means a_0=0. Therefore, $\frac{w_1}{w_2}=a_1r$ and $\frac{w_1}{w_2}=(\frac{1}{k})\frac{B(1)}{C(1)}$, so $\frac{B(1)}{C(1)}$ is equivalent to r. The genetic similarity (relatedness) between the recipient and the cooperative actors will be equivalent to the fitness similarity (relatedness) or the intrinsic fitness contribution ratio of a cooperative actor to recipient in linear evolution process.

The logic or mathematical process of reciprocity selection is the same with Hamilton's rule [47,51].